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ST 60xx OEM Hardware Guide

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TABLE OF CONTENTS

Contact and Legal Information 2
Contact Support 2
Trademark Notice 2
TABLE OF CONTENTS 3
List of Figures 5
List of Tables 6
Preface 7
Purpose 7
Notation 7
Reference 7
1 Overview 8
1.1 Key Features and Benefits 8
2 Compliance 9
3 Specifications 10
3.1 Connectors 10
3.1.1 Connector Details 10
3.1.2 Mating Connector Details 10
3.1.3 ST 60xx Connector 11
3.1.4 Mating Connector 11
3.1.5 Connector Electrical Pin Assignment 12
3.2 Antenna 13
3.3 Power 13
3.3.1 Input Power 13
3.3.2 Power Consumption 14
3.3.3 Inrush Current 14
3.4 Input/Output 15
3.4.1 Input Bandwidth 16
3.4.2 Output Bandwidth 16
3.5 Serial Interfaces 16
3.5.1 Console/Auxiliary (RS-232) Interface 16
3.6 Temperature Sensor 16
3.7 RF Specifications 17
3.7.1 Frequency 17
3.7.2 Antenna 17
3.8 GNSS 17
3.9 Accelerometer 18

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3.10 Mechanical 18
3.10.1 Dimensions 18
3.10.2 Weight 20
3.11 LED 20
3.12 Nonvolatile Storage 20
3.13 RAM 20
3.14 Environmental 21
3.14.1 Temperature 21
3.14.2 Environmental 21
4 Integration Guidelines 22
4.1 Enclosure Design 22
4.2 Ground Plane Requirements 22
4.2.1 Antenna Integration 23
4.3 Labeling 24

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LIST OF FIGURES

Figure 1: ST 60xx 8
Figure 2: ST 60xx Connector 11
Figure 3: Mating Connector 11
Figure 4: PIN Location 12
Figure 5: Onboard Antennas 13
Figure 6: Accelerometer Axis 18
Figure 7: Top View 19
Figure 8: Side View Dimensions (in.) 19
Figure 9: Bottom View Dimensions (in.) 20
Figure 10: Ground Plane Requirements (in.) 23
Figure 11: Antenna Integration Requirements (in.) 23

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LIST OF TABLES

Table 1: Connector Details 10
Table 2: Mating Connector Details 10
Table 3: Electrical Pin Assignment (ST 60xx) 12
Table 4: Input Power 14
Table 5: ST 60xx Power Consumption 14
Table 6: Inrush Current 15
Table 7: I/O Electrical Characteristics 15
Table 8: Analog Input Characteristics 15
Table 9: Electrical Characteristics 16
Table 10: Multi-GNSS Specifications 17

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PREFACE

Purpose

This document provides an overview of the hardware characteristics and specifications for the 20-pin ST 6000 and ST 6002.

Unless noted otherwise, the information in this guide applies to ST 6000 and ST 6002, hereinafter referred to as "ST 60xx"

Notation

An OEM Integrator is an ORBCOMM customer who purchases an ST 60xx for integration into their own enclosure. To become an OEM Integrator certain commercial criteria must be met. Contact your Account Executive for further details.

Hardware components and hardware stickers in this document might not be exactly as shown and are subject to change without notice.

CAUTION: This safety symbol warns of possible hazards to personnel, equipment, or both. It includes hazards that will or can cause personal injury, property damage, or death if the hazard is not avoided.

Note: A note indicates information with no potential hazard. A note indicates points of interest or provides supplementary information about a feature or task.

Numbered lists indicate a series of steps required to complete a task or function.

Bulleted lists highlight information where order or sequence is not crucial.

Reference

The content in the following Technical Publications may be useful in conjunction with this guide. These documents are available from the partner-support website, the ORBCOMM Developer Toolkit, or Support.

The names, numbers, and content of these documents are subject to change without notice. Always check the appropriate category (for example, ST Terminals) on the partner-support website for the most current version of each document.

[T404]	LSF Developer Guide
[T405]	LSF API Reference Guide

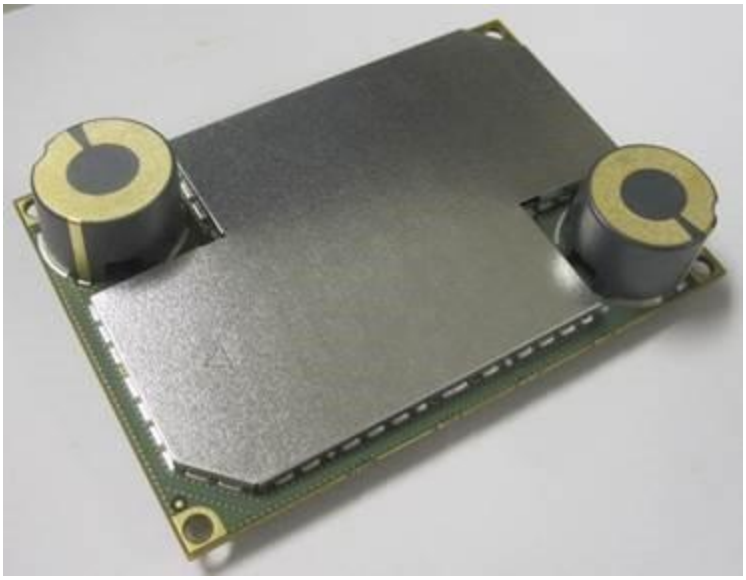
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1 OVERVIEW

The ST 60xx provides a high performance, low latency, two-way communication solution that uses ORBCOMM's satellite network.

The device consists of a Lua application controller, integral antennas, a satellite modem for communicating with the satellite, an integral GNSS subsystem, several input/output feeds capable of monitoring and controlling external sensors and devices, and also dedicated serial ports (two RS-232 ports). Solution Providers (SPs) must create a custom power supply for their particular application.

Figure 1: ST 60xx



The ST 60xx is suitable for both industrial and fixed applications, and it can work as a standalone data-messaging device, with built-in I/O data collection and processing capabilities. Feature-rich software tools make programming easy and shorten the design and testing time.

1.1 Key Features and Benefits

The ST 60xx has the following key features and benefits:

- Designed to be incorporated into a custom solution
- Built-in GNSS receiver to calculate position, speed, and heading that uses the satellite antenna
- Broad operational temperature range
- Satellite message payload and latency capabilities

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2 COMPLIANCE

The ST 60xx obtained the following certifications:

CE Mark

- EN 301 426

Inmarsat Type Approval

Industry Canada

- IC: RSS-170; ID: 11881A-ST6000

FCC Part 25

- FCC ID: XGS-ST6000
- CFR Title 47: Telecommunication, Part 25 - Satellite Communications, Sub-part C - Technical Standards
- OET 65 - Radiation Safety

RoHS

- Restriction of Hazardous Substances (RoHS) [1]

Mexico

- IFT TA Certificate

South Africa

- ICASA TA Certificate

[1] European Union's (EU) Directive 2002/95/EU "Restriction of Hazardous Substances" (RoHS) in Electronic and Electrical Equipment.

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3 SPECIFICATIONS

3.1 Connectors

3.1.1 Connector Details

See Table 1 for ST 60xx connector details.

Table 1: Connector Details

Parameter	
Manufacturer Part Number	Samtec SFC-110-T2-L-D-A-K-TR
Connector	Rectangular, shrouded female socket, SMT
Temperature Rating	-55°C to +125°C
Insulator Material	Black Liquid Crystal Polymer
Contact Material	Phosphor Bronze
Configuration	2 x 10, SMT connector
Contact Rating	3.1 A per pin

3.1.2 Mating Connector Details

The ST 60xx requires a mating connector (not included), see Table 2.

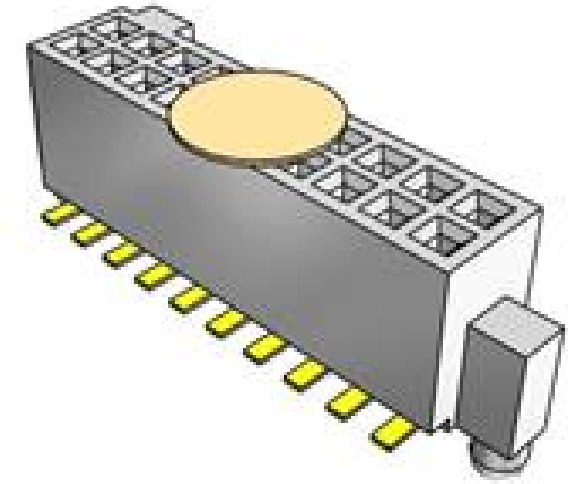
Table 2: Mating Connector Details

Parameter	
Manufacturer Part Number	Samtec TFC-110-32-L-D
Connector	Rectangular, shrouded male header, SMT
Temperature Rating	-55°C to +125°C
Insulator Material	Black Liquid Crystal Polymer
Device Material	Phosphor Bronze
Contact Rating	3.1 A per pin

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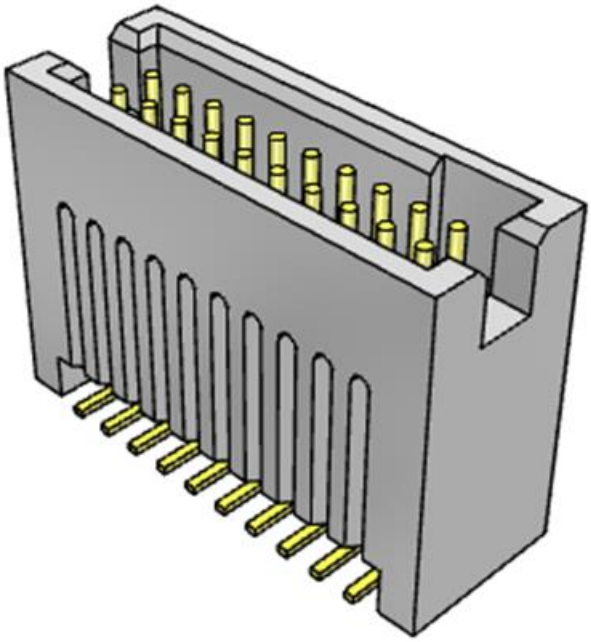
3.1.3 ST 60xx Connector

Figure 2: ST 60xx Connector



3.1.4 Mating Connector

Figure 3: Mating Connector



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3.1.5 Connector Electrical Pin Assignment

Figure 4: PIN Location

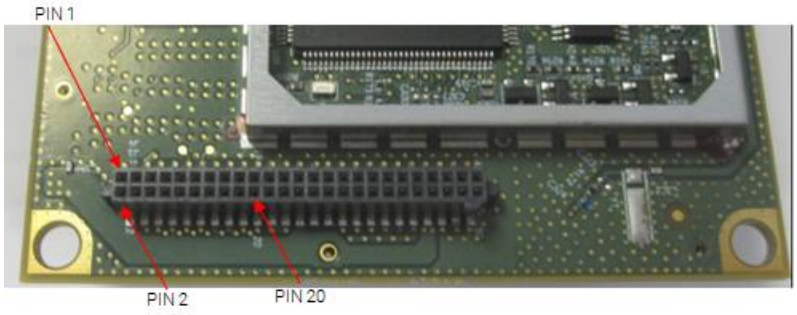


Table 3: Electrical Pin Assignment (ST 60xx)

PIN	Functionality	Description
1	GND	The negative terminal of the external power supply must be connected to this pin.
2	VIN	This pin is used to provide input power directly to the power amplifier circuitry. Regulated power must be applied to this pin whenever the device drives the TX_PWR EN line high and may be removed when the TX_PWR EN line is low. Alternatively, power may be applied to this pin continuously.
3	VAUX	This is the input power pin for the RF and baseband processors. Regulated power must be applied to this pin continuously to keep the device alive.
4	TX PWR EN	This is the control signal to apply power to the VIN pin. Power must be applied to VIN whenever this signal is high. If not used (VIN powered continuously), this pin should be left unconnected.
5	Reserved	-
6	AUX TX	This pin is 3.3 V compatible TTL signal to be connected to the receive input of the external serial port. May be used for logging and debugging.
7	Reserved	-
8	AUX RX	This pin is 3.3 V compatible TTL signal to be connected to the transmit output of the external serial port. May be used for logging and debugging.
9	Console TX	3.3V compatible TTL transmit output from the modem console port to be connected to the receive input of the external serial port.
10	OUT1	This pin is a 3.3 V TTL compatible connection for Output channel 5.
11	Console RX	This pin is 3.3 V compatible TTL receive input to the modem console port to be connected to the transmit output of the external serial port.
12	OUT2	This pin is a 3.3 V TTL compatible connection for Output channel 6.
13	I/O_1	This pin is 3.3V TTL compatible connection for Input / Output channel 1.
14	I/O_3	This pin is 3.3V TTL compatible connection for Input / Output channel 3.
15	I/O_2	This pin is 3.3V TTL compatible connection for Input / Output channel 2.
16	I/O_4	This pin is 3.3V TTL compatible connection for Input / Output channel 4.
17	1 PPS	This pin is 3.3V TTL compatible 1 pulse per second output generated by the internal GNSS module.

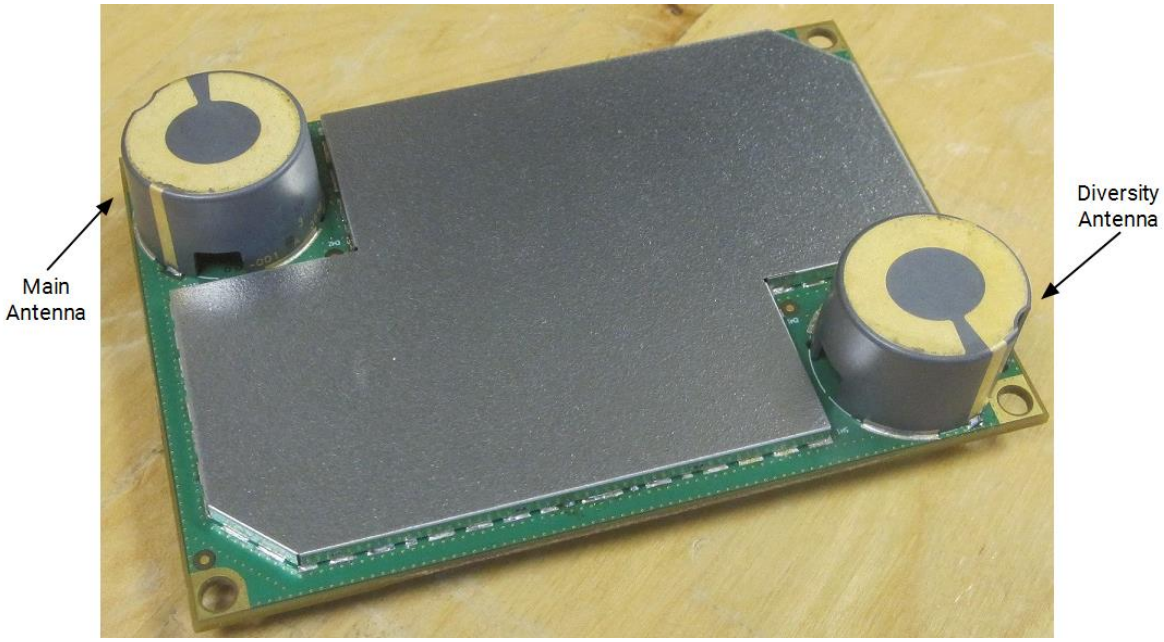
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PIN	Functionality	Description
18	Console Valid	This pin is 3.3V TTL compatible active high input that indicates if valid RS-232 levels are present on the external console receiver input.
19	AUX Valid	This pin is 3.3V TTL compatible active high input that indicates if valid RS-232 levels are present on the external auxiliary receiver input.
20	Reserved	-

3.2 Antenna

The ST 60xx supports two standard onboard antennas: Main and Diversity. Refer to [T406] for general antenna specifications.

Figure 5: Onboard Antennas



3.3 Power

3.3.1 Input Power

The ST 60xx has two power input pins (VAUX and VIN) and can be powered from either a single or dual external DC power source.

VAUX supplies power to the ST 60xx baseband and RF processors. Regulated power must be applied to this pin continuously to keep the device alive and synchronized with the satellite network.

VIN is the input power pin for the RF power amplifier circuitry. Regulated power must be applied to this pin whenever the device drives the TX_PWR EN line high and may be removed when the TX_PWR EN line is low. Alternatively, power may be applied to this pin continuously and TX_PWR_EN ignored.

Note: Input power protection is not provided on the ST 60xx.

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Table 4: Input Power

Parameter	Values
VAUX Voltage	3.5 to 5.8 VDC (6 V Max)
VAUX Ripple	50 mV p-p (max)
VAUX Current	600 mA (max)
VIN Voltage	5.8 VDC ± 3%
VIN Ripple	50 mV p-p (max)
VIN Current	1.6 A (max)
TX_PWR EN Asserted to VIN Stable	10 ms (max)
Reverse Polarity Protection	None
Over Voltage Protection	None
Over Current Protection	None
ESD Protection	Protection to ±30 kV contact discharge on all signals connected to the 20-pin interface header, except for the power pins.

3.3.2 Power Consumption

Typical power consumption values at VIN and VAUX=5.8 V and at room temperature (23°C)

Table 5: ST 60xx Power Consumption

Scenario	-40°C (-40°F)		RT		+85°C (+185°F)	
	VIN	VAUX	VIN	VAUX	VIN	VAUX
Shutdown	10 nA	7.93 µA	10 nA	11 µA	10 nA	32.1 µA
Sleep	10 nA	108 µA	10 nA	129 µA	10 nA	211 µA
Power On	18.2 µA	23 mA	20.2 µA	24 mA	22.4 µA	26 mA
RFIC On	18.2 µA	37 mA	19.5 µA	38 mA	22.1 µA	40 ma
RFIC + GPS On	18.2 µA	84 mA	19.3 µA	83 mA	21.9 µA	84 mA
Tx On	0.83 A	275 mA	0.85 A	273 mA	0.81 A	274 mA

3.3.3 Inrush Current

Inrush current transients occur when the ST 60xx:

- First powers on
- Wakes up
- Applies power to the RF transmit circuitry

These transients are of very short duration and low energy but can have high peak currents. The power supply must be capable of handling these inrush currents without dipping below the minimum input voltage of the respective power rails. Typical inrush currents are listed in the following table.

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Table 6: Inrush Current

Scenario[1]	-40°C (-40°F)		RT		+85°C (+185°F)	
	VIN	VAUX	VIN	VAUX	VIN	VAUX
Power Application	-	1.71 A (<343µs)	-	1.9 A (<263µs)	-	1.6 A (<263µs)
RFIC On	-	1.4 A (<333µs)	-	2 A (<243µs)	-	1.7 A (<253µs)
GPS On	-	1 A (<243µs)	-	1 A (<173µs)	-	1 A (<212µs)
Tx Activation (with main antenna)	1.5 A (<1ms)	0.7 A (<262µs)	1.7 A (<1ms)	0.6 A (<105µs)	0.85 A (<1ms)	0.35 A (<425µs)

3.4 Input/Output

The ST 60xx I/Os are each independently operable and configurable in one of the following modes:

- Analog input
- Digital input
- Digital output – push pull

The ST 60xx is equipped with various I/O ports. Refer to the pin number descriptions in section 3.1.5 for an explanation of each I/O signal. To connect these ports to an external device you may need to use a suitable translator depending on what the signal is connected to.

The I/O electrical characteristics of this interface are shown in Table 7 and Table 8 shows the analog characteristics.

Table 7: I/O Electrical Characteristics

Parameter	Min.	Typ.	Max.	Units
Input high voltage threshold	2.31	-	-	V
Input low voltage threshold	-	-	1.155	V
Output high voltage threshold	2.8	-	-	V
Output low voltage threshold	-	-	0.5	V
Output high voltage, high drive strength	-	-	-8	mA
Output high voltage, low drive strength	-	-	-2	mA
Output low voltage, high drive strength	-	-	9	mA
Output low voltage, low drive strength	-	-	2	mA

OUT1 and OUT2 are dedicated push-pull outputs only. These outputs may be used to drive off board circuitry capable of driving high current loads.

Table 8: Analog Input Characteristics

Parameter	Min.	Typ.	Max.	Units
Input resistance	-	2	5	kΩ

[1]All VIN and VAUX measurements were performed at 5.8 V.

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Parameter	Min.	Typ.	Max.	Units
Input capacitance	-	4	5	pF
Normal input range	0	-	3.3	V
Resolution (12 bits)	-	0.8	-	mV
Full scale error	-	-	5.4	LSB

3.4.1 Input Bandwidth

When used as a digital or analog input, the bandwidth of the I/O circuitry is 1 kHz or more.

Note: This does not imply that the software must have a sample rate greater than 1 Hz.

3.4.2 Output Bandwidth

When used as a digital output the I/O circuitry has a bandwidth exceeding 100Hz.

Note: This does not imply that the software must have the capability of generating high rate pulse trains at frequencies > 10 Hz.

3.5 Serial Interfaces

The electrical characteristics (Table 9) of the interface are:

Table 9: Electrical Characteristics

Parameter	Min.	Typical	Max.	Units
Rx input high voltage threshold	2.31			V
Rx input low voltage threshold			1.155	V
Tx output high voltage threshold	2.8			V
Tx output low voltage threshold		-	0.5	V

3.5.1 Console/Auxiliary (RS-232) Interface

The ST 60xx is equipped with an RS-232 console and an auxiliary TTL level serial port. The auxiliary port is primarily intended for debug output or software upgrade downloads. The console and auxiliary serial port is connected directly to the UART of the microcontroller and therefore communicates via CMOS level RS-232 style signaling (0-3.3 V, non-inverted).

The console and auxiliary serial interfaces to the ST 60xx are asynchronous with 1 start bit, 8 data bits, no parity, 1 stop bit; the default rate is 9600 bps, while the maximum rate is 115.2 kbps. The interfaces are 3.3 V TTL compatible.

3.6 Temperature Sensor

Parameter	Value
Temperature reading range	-40° to +85°C (-40° to +185°F)
Temperature reading accuracy	± 0.5 degrees typical, ± 3.0 degrees maximum

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3.7 RF Specifications

3.7.1 Frequency

Parameter	OGx Value	IsatData Pro Value
Receive		
Frequency Band	1525 to 1559 MHz	1525 to 1559 MHz
Modulation	OQPSK	OQPSK
Symbol Rate	2000, 8000, 16000 symbols/seconds	3000 symbols/second
Polarization	RHCP	RHCP
Transmit		
Frequency Band	1626.5 to 1660.5 MHz	1626.5 to 1660.5 MHz
Modulation	OQPSK	OQPSK
Symbol Rate	800, 1600, 3200, 6400 12800 symbols/seconds	900 symbols/second (maximum)
Polarization	RHCP	RHCP

3.7.2 Antenna

Parameter	Value
Maximum EIRP	7 dBW
Elevation Angle	0 to 90 degrees
Maximum transmit antenna gain	4.2 dBic

3.8 GNSS

The manufacturer's specifications are given in the table below.

Table 10: Multi-GNSS Specifications

Parameter	GPS	GLONASS	BeiDou	GPS/GLONASS	GPS/BeiDou
Time to First Fix[1]					
Cold Start	29 s	30 s	36 s	26 s	27 s
Hot Start	1 s	1 s	1 s	1 s	1 s
Sensitivity					
Tracking	-166 dBm	-166 dBm	-160 dBm	-167 dBm	-165 dBm
Hot Start	-156 dBm	-155 dBm	-155 dBm	-156 dBm	-156 dBm
Cold Start	-148 dBm	-145 dBm	-138 dBm	-148 dBm	-148 dBm
Accuracy					

[1] All satellites at -130 dBm

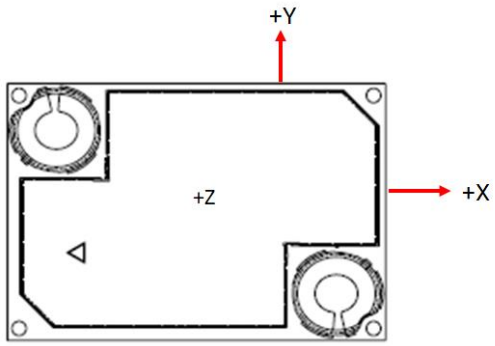
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Parameter	GPS	GLONASS	BeiDou	GPS/GLONASS	GPS/BeiDou
Horizontal Position (CEP) [1]	2.5/2.0 m	4.0 m	3.0 m	2.5/2.0 m	2.5/2.0 m
Velocity	0.05 m/s	0.1 m/s	0.1 m/s	0.05 m/s	0.05 m/s
Heading	0.3 degrees	0.4 degrees	-	0.3 degrees	0.3 degrees

3.9 Accelerometer

The device has an internal 3D accelerometer to detect motion in any axis. Figure 6 shows the positive axes for the accelerometer.

Figure 6: Accelerometer Axis



The accelerometer is very important for low power devices when it is critical to save power while stationary and quickly detect when motion starts. In powered devices where low power is not critical, GPS can be polled to detect motion. However, for low power applications frequent GPS fixes can dominate the power budget. To reduce power budget effects of GPS fixes, an accelerometer can be used to trigger a GPS fix to detect if motion occurred.

The accelerometer thresholds to detect motion vary depending on the environment. In order to avoid, false motion detects, extensive testing is required to ensure that adequate acceleration magnitude thresholds and time durations are used. Refer to [T405] for all accelerometer specifications.

If using the accelerometer, mount the device in one of the configurations shown in [T405].

3.10 Mechanical

3.10.1 Dimensions

	Length	Width	Height
Without connectors (JTAG or strip)	10.1 cm / 3.97 in	6.9 cm / 2.7 in	1.0 cm / 0.39 in
With antennas			2.1cm/0.83 in

[1] CEP, 50%, 24 hours static, -130 dBm

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Figure 7: Top View

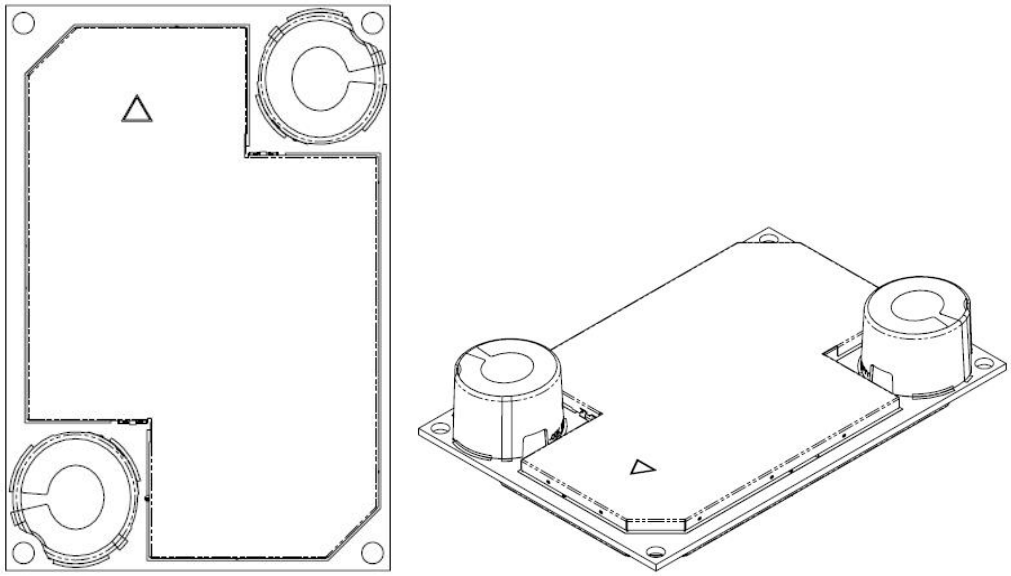


Figure 8: Side View Dimensions (in.)

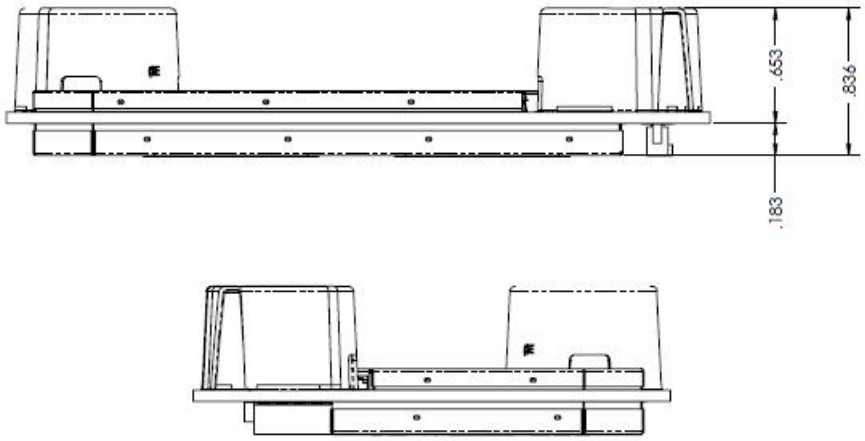
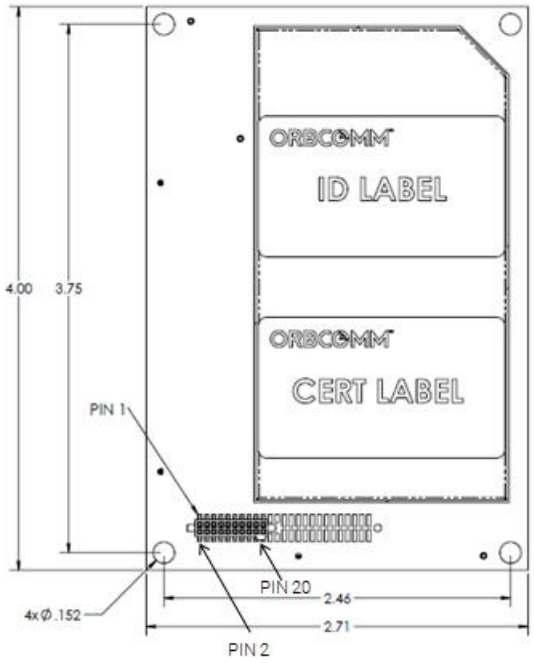


Figure 9: Bottom View Dimensions (in.)



3.10.2 Weight

Parameter	Value
Weight	59.4 g

3.11 LED

The ST 60xx has a red visual indicator that is controlled by the software. It is a power indicator during initial power on, otherwise it is off.

3.12 Nonvolatile Storage

The device has nonvolatile flash memory, which is shared by both the device firmware and user services.

Parameter	Value
Nonvolatile Onboard Flash Storage	ST 6000 - 8 MB ST 6002 - 16 MB
Write-Erase Cycles (per operating life)	100 000

3.13 RAM

Parameter	Value
RAM (volatile memory)	4 MB

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3.14 Environmental

3.14.1 Temperature

Parameter	Value
Operating Temperature	-40° to +85°C (-40° to +185°F)
Storage Temperature	-40° to +85°C (-40° to +185°F)

3.14.2 Environmental

Parameter	Description
Vibration	The ST 60xx meets all its specifications during exposure to random vehicular vibration levels per SAE J1455, section 4.9.4.2 figures 6, 7, and 8, and MIL-STD-810G, section 514.6, figure 514.6C-1.
Mechanical Shock	The ST 60xx meets all its specifications after exposure to positive and negative saw tooth shock pulses with peaks of 20G and durations of 11 ms as specified in MIL-STD-810G, section 516.6, Procedure I, section 2.3.2c.
Altitude	The ST 60xx meets all of its specifications after a non-operating 12.2 km altitude test as detailed in SAE J1455, section 4.9.3, except with an ambient temperature of -40 °C.
Thermal Shock	The ST 60xx meets all of its specifications after a thermal shock test as detailed in SAE J1455, section 4.1.3.2.
ESD	The ST 60xx meets all its specifications after exposure of the ST 60xx to 2 kV ESD contact discharge per IEC 60945 and IEC 61000-4-2, level 3. All the connections on the connectors, except for the VIN and VAUX power rails, are ESD protected to ±30 kV contact discharge according to IEC 61000-4-2 far exceeding level 4.

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4 INTEGRATION GUIDELINES

This section contains a number of guidelines to assist the Solution Provider (SP) in building their ST 60xx enclosure. It must be recognized that this section provides guidelines only and each SP must use their own discretion to finalize the integration approach that works for them.

4.1 Enclosure Design

The ST 60xx is not designed for outdoor environments. Consequently, the ST 60xx requires a robust environmentally sealed enclosure that can house the ST 60xx.

The following guidelines are recommended for the enclosure design.

- An IP67 rating or better for outdoor use.
- Use enclosure materials that are transparent to L-Band (1-2 GHz) radio signals.
- Two recommended enclosure materials are:
    - Xenoy® Resin 5220U. This plastic material offers good chemical and UV resistance and great impact resistance even at low temperatures.
    - Lexan EXL 9330.

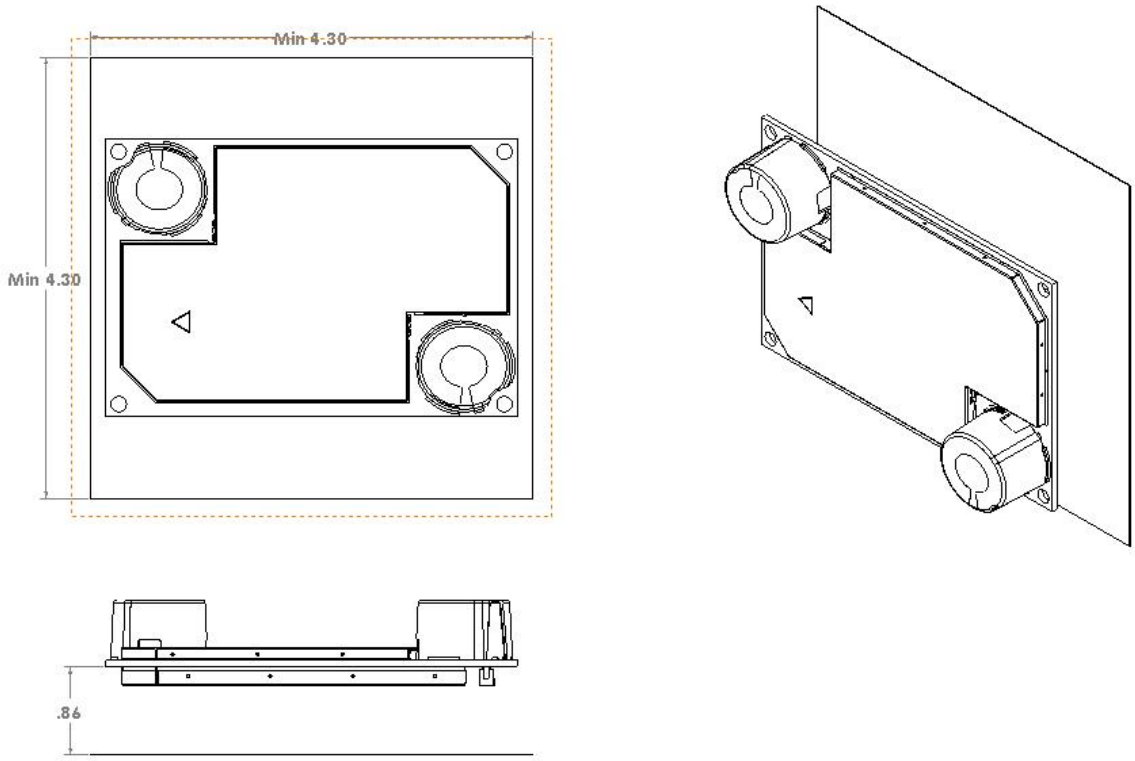
4.2 Ground Plane Requirements

A metal plate is required under the antenna board. The optimal recommended separation between the metal plate and the modem card is 0.86”. This can be reduced to 0.5” with some moderate impact on low elevation gain.

If the ST 60xx is installed on a metal surface larger than the metal plate size indicated in Figure 10, then no metal plate is needed as long as the surface-to-antenna distance is the same as shown in Figure 10.

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Figure 10: Ground Plane Requirements (in.)

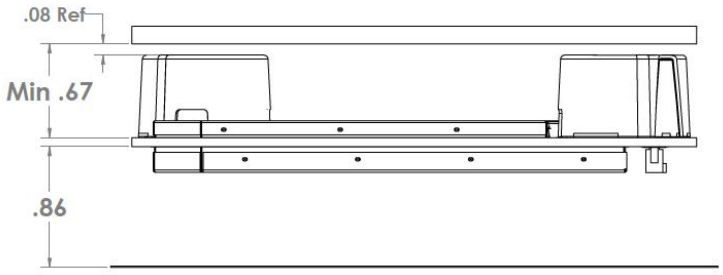


4.2.1 Antenna Integration

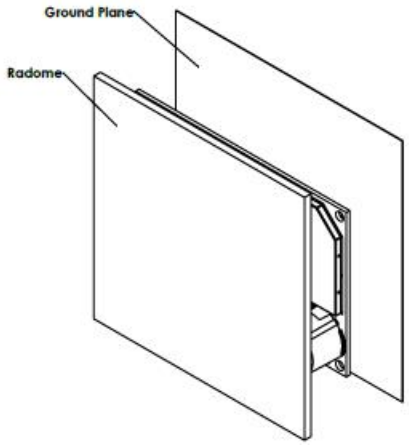
The minimum distance (air gap) recommended between the antenna and the enclosure is 2 mm.

Note: Antennas are fragile elements that can withstand vibration but not direct shock or contact.

Figure 11: Antenna Integration Requirements (in.)



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4.3 Labeling

The mobile ID on the ST 60xx is the network identification number. It is recommended that the SP place a copy of the mobile ID on the exterior of the enclosure housing the device. With the mobile ID on the enclosure, installers can readily identify the network identification number.

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